

Prof. Anand Kumar Jha

Research publication portfolio

Position	Professor, Department of Physics
Affiliation	Indian Institute of Technology Kanpur, India
Research areas	quantum optics; optical coherence; quantum entanglement
Scholar metrics	2355 citations; h-index 21; i10-index 33
Coverage	96 profile records, compiled to 77 distinct works, 2005–2026
Profile	Google Scholar record

Year-wise publication matrix

Rows are shaded by complete year blocks. Position is the researcher’s place in the abbreviated author list (position/total). Click an arXiv number to open the PDF.

Year	Pos.	Authors (abbrev.)	Paper title	Preprint posted	Cited	Journal / publication	arXiv PDF
2026	3/3	N. Senapati; A. Bhattacharjee; A. K. Jha	Scattering-robust Imaging of Azimuthal Features with Enhanced Resolution	2026-09-10	–	arXiv preprint	2609.11113
2026	4/4	P. Mohta; S. Aarav; H. Defienne; A. K. Jha	Spatial coherence enabled sensorless adaptive optical imaging	2026-08-12	–	arXiv preprint	2608.11800
2026	4/4	H. Jain; S. Karan; R. Prasad; A. K. Jha	Non-monotonic dependence of OAM Schmidt spectrum on crystal thickness	2026-08-07	–	arXiv preprint	2608.07090
2026	4/4	S. Karan; S. Chaudhary; H. Wanare; A. K. Jha	OAM-mode sorting with a wavefront twister: a proposal	2026-04-30	–	Journal of Optics 28 (8), 085202	2604.27739
2026	4/4	R. Prasad; N. Senapati; A. Bhattacharjee; A. K. Jha	Radial Schmidt mode detector of entangled photons	2026-06-24	–	Journal of Optics	2606.25735
2026	5/5	R. Prasad; N. Senapati; A. Bhattacharjee; S. Karan; A. K. Jha	Measuring the complete set of spatial Schmidt modes of entangled two-photon fields	2026-05-18	–	arXiv preprint	2605.17974
2026	4/4	N. Senapati; A. Bhattacharjee; K. Khare; A. K. Jha	Super-resolution imaging of azimuthal features with illumination carrying OAM	2025-11-12	2	Applied Physics Letters 128 (18)	2511.09066
2026	4/4	S. Karan; A. Srivastava; P. S. Todkar; A. K. Jha	Polarization-preserving wavefront rotator	2026-04-28	–	arXiv preprint	2604.25475
2026	2/2	S. Karan; A. K. Jha	Observation of OAM non-conservation in entangled photon generation	2026-04-26	–	arXiv preprint	2604.23550
2026	5/5	P. Mohta; K. Moliya; A. Nag; S. Aarav; A. K. Jha	Noise-resilient imaging through coherence filtering	2026-05-10	2	Physical Review Applied 25 (3), 034001	2605.09324
2026	3/3	N. Senapati; G. Kulkarni; A. K. Jha	Theory of temporal three-photon interference	2026-02-17	–	arXiv preprint	2602.15573

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Year	Pos.	Authors (abbrev.)	Paper title	Preprint posted	Cited	Journal / publication	arXiv PDF
2025	3/3	P. Mohta; A. Bhattacharjee; A. K. Jha	Complex spatial coherence measurement using phase-shifting wavefront-inversion interferometry	–	2	Journal of Optics 27 (7), 075606	–
2025	3/3	S. Karan; M. P. van Exter; A. K. Jha	Broadband uniform-efficiency OAM-mode detector	2025-02-20	13	Science Advances 11 (11), eadq7201	2502.14711
2025	7/7	R. Prasad; N. Senapati; S. Karan; A. Bhattacharjee; B. Piccirillo; M. A. Alonso; A. K. Jha	Experimental technique for measuring radial coherence	2025-02-22	3	Optics Express 33 (5), 11693–11701	2502.16063
2024	6/6	R. Prasad; S. Wanare; S. Karan; M. K. Joshi; A. Bhattacharjee; A. K. Jha	Structured position-momentum-entangled two-photon fields	2024-12-14	7	Physical Review Applied 22 (6), 064034	2412.10954
2024	6/6	R. Prasad; N. Senapati; A. Bhattacharjee; B. Piccirillo; M. A. Alonso; A. K. Jha	Measuring Radial Coherence	–	–	CLEO/Pacific Rim, Mo4H_4	–
2024	3/3	S. Karan; N. Senapati; A. K. Jha	Wavefront rotator with near-zero mean polarization change	2024-05-17	3	Applied Optics 63 (17), 4552–4557	2405.10804
2024	3/3	P. Mohta; A. Bhattacharjee; A. K. Jha	Eight-shot measurement of spatially non-stationary complex coherence function	2024-02-29	1	arXiv preprint	2403.00090
2023	3/3	S. Karan; R. Prasad; A. K. Jha	Postselection-free controlled generation of a high-dimensional orbital-angular-momentum entangled state	2022-03-28	18	Physical Review Applied 20 (5), 054027	2203.14799
2023	6/6	R. Prasad; S. Wanare; S. Karan; M. K. Joshi; A. Bhattacharjee; A. K. Jha	Structuring two-photon correlation of entangled field: experimental demonstration	–	–	Laser Science, LM5F.5	–
2023	3/3	S. Karan; M. P. van Exter; A. K. Jha	Broadband uniform-efficiency OAM detector	2025-02-20	–	Laser Science, JTu5A.15	2502.14711
2023	6/6	R. Prasad; S. Wanare; S. Karan; M. K. Joshi; A. Bhattacharjee; A. K. Jha	Structured near-field spatial correlations in entangled photons	–	–	Quantum 2.0, QTh2A.26	–
2023	4/4	A. Bhattacharjee; S. Biswas; M. A. Alonso; A. K. Jha	Coherence in the radial degree of freedom	–	5	Journal of the Optical Society of America A 40 (3), 411–416	–
2022	4/4	S. Karan; Ruchi; P. Mohta; A. K. Jha	Quantifying polarization changes induced by rotating Dove prisms and K-mirrors	2022-08-01	13	Applied Optics 61 (28), 8302–8307	2208.00828
2022	5/5	A. Bhattacharjee; M. K. Joshi; S. Karan; J. Leach; A. K. Jha	Propagation-induced revival of entanglement in the angle-OAM bases	2021-11-08	32	Science Advances 8 (31), eabn7876	2111.04420
2022	3/3	N. Meher; M. Bhattacharya; A. K. Jha	Direct measurement of atomic entanglement via cavity photon statistics	2022-04-08	2	Annalen der Physik 534 (6), 2100395	2204.03891
2022	3/3	A. Bhattacharjee; N. Meher; A. K. Jha	Measurement of two-photon position–momentum Einstein–Podolsky–Rosen correlations through single-photon intensity measurements	2021-02-08	18	New Journal of Physics 24 (5), 053033	2102.04356
2021	6/7	P. Kumar; T. Biswas; K. Feliz; R. Kanamoto; M. S. Chang; A. K. Jha; M. Bhattacharya	Cavity optomechanical sensing and manipulation of an atomic persistent current	–	38	Physical Review Letters 127 (11), 113601	–

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Year	Pos.	Authors (abbrev.)	Paper title	Preprint posted	Cited	Journal / publication	arXiv PDF
2020	2/2	N. Meher; A. K. Jha	Photon statistics in parametric down-conversion process	–	1	Frontiers in Optics, JTu1B.38	–
2020	4/4	A. Bhattacharjee; M. K. Joshi; J. Leach; A. K. Jha	Realization of strong far-field two-photon correlation in a transverse degree of freedom	–	–	Laser Science, JTu1A.23	–
2020	3/3	G. Kulkarni; S. Karan; A. K. Jha	Measuring pure states in the orbital angular momentum basis using nine image acquisitions	2020-05-08	–	Frontiers in Optics, FM7C.1	2005.03849
2020	8/8	S. Karan; S. Aarav; H. Bharadhwaj; L. Taneja; A. De; G. Kulkarni; N. Meher; A. K. Jha	Phase matching in β -barium borate crystals for spontaneous parametric down-conversion	2018-10-02	115	Journal of Optics 22 (8), 083501	1810.01184
2020	2/2	A. Bhattacharjee; A. K. Jha	Experimental demonstration of structural robustness of spatially partially coherent fields in turbulence	2020-06-21	15	Optics Letters 45 (14), 4068–4071	2006.11724
2020	2/2	N. Meher; A. K. Jha	Dependence of the photon statistics of down-converted field-modes on the photon statistics of the pump field-mode	2020-06-19	6	Journal of the Optical Society of America B 37 (8), 2248–2255	2006.12271
2020	2/2	A. Bhattacharjee; A. K. Jha	Measurement of two-photon position-momentum EPR-correlation by detecting single-photon correlation	2021-02-08	–	CLEO: Applications and Technology, JTu2A.3	2102.04356
2020	3/3	G. Kulkarni; S. Karan; A. K. Jha	Measurement of pure states of light in the orbital-angular-momentum basis using nine multipixel image acquisitions	2020-05-08	10	Physical Review Applied 13 (5), 054077	2005.03849
2020	4/4	A. Bhattacharjee; S. Aarav; H. Wanare; A. K. Jha	Controlling propagation of spatial coherence for enhanced imaging through scattering media	2018-10-01	7	Physical Review A 101 (4), 043839	1810.00686
2020	4/4	N. Meher; A. S. M. Patoary; G. Kulkarni; A. K. Jha	Intrinsic degree of coherence of two-qubit states and measures of two-particle quantum correlations	2020-03-06	12	Journal of the Optical Society of America B 37 (4), 1224–1230	2003.03372
2019	3/3	A. Bhattacharjee; R. Sahu; A. K. Jha	Generation of a Gaussian Schell-model field as a mixture of its coherent modes	2018-08-27	21	Journal of Optics 21 (10), 105601	1808.08774
2019	3/3	A. S. M. Patoary; G. Kulkarni; A. K. Jha	Intrinsic degree of coherence of classical and quantum states	2017-12-10	17	Journal of the Optical Society of America B 36 (10), 2765–2776	1712.03475
2018	3/3	A. Bhattacharjee; S. Aarav; A. K. Jha	Two-shot measurement of spatial coherence	2017-12-11	21	Applied Physics Letters 113 (5)	1712.04517
2018	4/4	G. Kulkarni; L. Taneja; S. Aarav; A. K. Jha	Angular Schmidt spectrum of entangled photons: derivation of an exact formula and experimental characterization for noncollinear phase matching	2017-12-09	25	Physical Review A 97 (6), 063846	1712.03355
2018	6/6	R. Sahu; S. Chaudhary; K. Khare; M. Bhattacharya; H. Wanare; A. K. Jha	Angular lens	2017-11-09	55	Optics Express 26 (7), 8709–8718	1711.03266
2017	2/2	G. Kulkarni; A. K. Jha	Polarization coherence and entanglement	–	–	Selected Topics in Photonics, 61–67	–
2017	5/5	G. Kulkarni; R. Sahu; O. S. Magaña-Loaiza; R. W. Boyd; A. K. Jha	Single-shot measurement of the orbital-angular-momentum spectrum of light	2016-12-01	122	Nature Communications 8 (1), 1054	1612.00335
2017	4/4	S. Aarav; A. Bhattacharjee; H. Wanare; A. K. Jha	Efficient generation of propagation-invariant spatially stationary partially coherent fields	2017-07-21	17	Physical Review A 96 (3), 033815	1707.06850
2017	3/3	G. Kulkarni; P. Kumar; A. K. Jha	Transfer of temporal coherence in parametric down-conversion	2017-04-06	23	Journal of the Optical Society of America B 34 (8), 1637	1704.01786

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2017	5/5	M. Bhattacharya; B. Rodenburg; W. Wetzel; B. Ek; A. K. Jha	Effects of photon scattering torque in off-axis levitated torsional cavity optomechanics	–	1	Journal of the Optical Society of America B 34 (6), C44–C51	–
2017	4/5	W. Wetzel; B. Rodenburg; B. Ek; A. K. Jha; M. Bhattacharya	Rotational cavity optomechanics	–	–	APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts	–
2016	1/1	A. K. Jha	Transfer of quantum correlations from one-photon systems to entangled two-photon systems	–	–	International Conference on Fibre Optics and Photonics, Tu2C.1	–
2016	3/3	G. Kulkarni; V. Subrahmanyam; A. K. Jha	Intrinsic upper bound on two-qubit polarization entanglement predetermined by pump polarization correlations in parametric down-conversion	2015-11-30	18	Physical Review A 93 (6), 063842	1511.09288
2015	3/5	M. Singh; K. Khare; A. K. Jha; S. Prabhakar; R. P. Singh	Accurate multipixel phase measurement with classical-light interferometry	2014-12-10	24	Physical Review A 91 (2), 021802	1412.3207
2011	1/3	A. K. Jha; G. S. Agarwal; R. W. Boyd	Partial angular coherence and the angular Schmidt spectrum of entangled two-photon fields	–	53	Physical Review A 84 (6), 063847	–
2011	2/6+	R. W. Boyd; H. Shin; M. Malik; C. O’Sullivan; K. W. C. Chan; H. J. Chang; et al.	Applications of nonlinear optics in quantum imaging and quantum communication	–	3	Nonlinear Optics: Materials, Fundamentals and Applications, NWC2	–
2011	16/16	S. M. Barnett; A. M. Yao; F. M. Miatto; L. Allen; D. T. Pegg; J. Götze; M. J. Padgett; et al.	Quantum description of the angular coordinate and angular momentum	–	–	International Conference on Quantum Information, QTuE1	–
2011	1/3	A. K. Jha; G. S. Agarwal; R. W. Boyd	Supersensitive measurement of angular displacements using entangled photons	2010-11-12	113	Physical Review A 83 (5), 053829	1011.2833
2011	2/6	R. W. Boyd; A. K. Jha; M. Malik; C. O’Sullivan; B. Rodenburg; D. J. Gauthier	Quantum key distribution in a high-dimensional state space: exploiting the transverse degree of freedom of the photon	–	56	Advances in Photonics of Quantum Computing, Memory, and Communication IV	–
2010	5/10	R. W. Boyd; J. Leach; B. Jack; J. Romero; A. K. Jha; A. M. Yao; S. Franke-Arnold; et al.	New twist on light beams for quantum information science	–	5	Optics and Photonics News 21 (12), 48	–
2010	4/10	J. Leach; B. Jack; J. Romero; A. K. Jha; A. M. Yao; S. Franke-Arnold; D. G. Ireland; et al.	Quantum correlations in optical angle-orbital angular momentum variables	–	756	Science 329 (5992), 662–665	–
2010	1/3	A. K. Jha; G. A. Tyler; R. W. Boyd	Effects of atmospheric turbulence on the entanglement of spatial two-qubit states	–	74	Physical Review A 81 (5), 053832	–
2010	1/7	A. K. Jha; J. Leach; B. Jack; S. Franke-Arnold; S. M. Barnett; R. W. Boyd; M. J. Padgett	Angular two-photon interference and angular two-qubit states	–	61	Physical Review Letters 104 (1), 010501	–
2010	2/2	A. K. Jha; R. W. Boyd	Spatial two-photon coherence of the entangled field produced by down-conversion using a partially spatially coherent pump beam	–	78	Physical Review A 81 (1), 013828	–

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2009	5/6+	J. Leach; B. Jack; J. Romero; R. W. Boyd; A. K. Jha; S. M. Barnett; et al.	Continuous variable EPR paradox for angle and orbital angular momentum	–	–	Laser Science, JWD4	–
2009	6/9	J. Leach; B. Jack; J. Romero; M. Ritsch-Martel; R. W. Boyd; A. K. Jha; S. M. Barnett; et al.	Violation of a Bell inequality in two-dimensional orbital angular momentum state-spaces	–	248	Optics Express 17 (10), 8287–8293	–
2009	3/7	R. W. Boyd; K. W. C. Chan; A. K. Jha; M. Malik; C. O’Sullivan; H. Shin; P. Zerom	Quantum imaging: enhanced image formation using quantum states of light	–	11	Quantum Information and Computation VII 7342, 88–90	–
2009	1/1	A. K. Jha	Coherence properties of the entangled two-photon field produced by parametric down conversion	–	8	University of Rochester (PhD thesis)	–
2008	1/3	A. K. Jha; M. Malik; R. W. Boyd	Exploring energy-time entanglement using geometric phase	2017-11-14	52	Physical Review Letters 101 (18), 180405	1711.05079
2008	1/8+	A. K. Jha; B. Jack; E. Yao; J. Leach; R. W. Boyd; G. S. Buller; S. M. Barnett; et al.	Fourier relationship between angular position and orbital angular momentum of entangled photons	–	–	Laser Science, LThE2	–
2008	1/9	A. K. Jha; B. Jack; E. Yao; J. Leach; R. W. Boyd; G. S. Buller; S. M. Barnett; S. Franke-Arnold; M. J. Padgett	Fourier relationship between the angle and angular momentum of entangled photons	–	86	Physical Review A 78 (4), 043810	–
2008	4/5	R. W. Boyd; G. S. Agarwal; K. W. C. Chan; A. K. Jha; M. N. O’Sullivan	Propagation of quantum states of light through absorbing and amplifying media	–	25	Optics Communications 281 (14), 3732–3738	–
2008	1/4	A. K. Jha; M. N. O’Sullivan; K. W. C. Chan; R. W. Boyd	Temporal coherence and indistinguishability in two-photon interference effects	2007-12-17	47	Physical Review A 77 (2), 021801	0712.2735
2007	2/7	S. G. Lukishova; A. K. Jha; N. Savidis; S. White; L. Bissell; L. Elgin; C. R. Stroud	Quantum optics teaching laboratory	–	2	Conference on Coherence and Quantum Optics, JWC8	–
2007	3/5	K. W. C. Chan; M. N. O’Sullivan-Hale; A. K. Jha; R. W. Boyd; G. A. Tyler	Effects of turbulence on the transverse position-momentum entanglement of biphotons	–	–	2007 Quantum Electronics and Laser Science Conference, 1–2	–
2006	2/6	G. Selvaraj; A. K. Jha; P. Lougovski; M. Florescu; R. W. Boyd; J. P. Dowling	Single photon source using chiral nematic liquid crystal	–	–	APS March Meeting Abstracts, V40.002	–
2005	3/5	S. G. Lukishova; C. R. Stroud; A. K. Jha; L. Elgin; S. Schrauth	Quantum optics and quantum information teaching laboratory at the Institute of Optics, University of Rochester	–	2	Frontiers in Optics, OSA Technical Digest Series	–
–	3/3	A. Bhattacharjee; S. Aarav; A. K. Jha	Depth free imaging through scattering medium	–	–	undated profile record	–
–	1/3	A. K. Jha; G. S. Agarwal; R. W. Boyd	Orbital angular momentum: interference, entanglement, and precision measurement	–	–	undated profile record	–

Reading notes. Alternating pale-blue and white bands apply to complete profile-year groups. Author position is the researcher’s place in the abbreviated author list. Where Google Scholar truncated a list, the full roster was recovered from the publisher record, so those positions are exact; the three lists still carrying a trailing + are conference records with no publisher roster, where the true total is at least the number shown. The pale shading marks the 2026, 2024, 2022, 2020, 2018, 2016, 2011, 2009, 2007 and 2005 groups.

Deduplication. The profile carries 96 records, of which 89 belong to this researcher; the remaining 7 are name collisions with other people named A. K. Jha — a pharmacology trial, an ethnobotany paper, two finance and management studies, an agronomy paper and a petroleum-engineering report — and have been left out. Of the 89 genuine records, 12 are duplicate or garbled re-listings of the same work (for example, one 2019 paper appears four times under three different titles, and one 2008 paper twice under two venue spellings); these are merged so that each work appears once, at its journal version where one exists. The result is the 77 distinct works tabulated here. Titles are reproduced from the profile with capitalization normalised.

Preprint column. This gives the first arXiv posting date, not a journal received date. Several papers were posted to arXiv long after they appeared in print — most visibly the 2008 *Physical Review Letters* paper whose preprint dates from 2017 — so a preprint date later than the publication year is genuine and not an error. Optics and photonics venues such as *Optics Express*, *Applied Optics* and *Journal of the Optical Society of America* frequently carry no preprint at all, and those cells are dashed.

Citation column. Counts are as displayed on the profile and are dashed where the profile shows none, which is normal for conference abstracts, book chapters and recent preprints.

Undated records. Two profile records carry no year and no venue on Google Scholar; they are listed at the foot of the table rather than assigned a guessed year.

Data snapshot. Public Google Scholar profile, checked 27 September 2026. Scholar metrics are dynamic and may change after this snapshot.